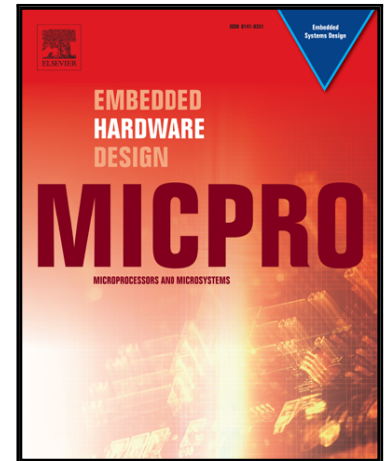


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Design and Implementation of ANN-based SAPF Approach for Current Harmonics Mitigation in Industrial Power Systems

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Abstract

This article describes with Shunt Active Power Filter (SAPF) for compensation of reactive Power (P), current harmonics. An ADALINE based neural network current estimator can control the SAPF to maintain sinusoidal source currents and unity Power Factor (PF). The three-phase (III-Phase) load currents are sensed and measured by Least Mean Square(LMS) algorithm based on ADALINE, the weights obtained with the help of online calculation, and then these weights are multiplied with the help of unit vector, which gives the fundamental frequency of Load Currents (LC). The DC link voltage (V) of the V Source Converter, maintained at a constant value, and it can work as the SAPF, and various simulation results are presented under the steady-state conditions using the PI controller. The switching action of VSC is performed using hysteresis current control based on pulse width modulation, which makes the source currents should follow the reference source currents. The MATLAB based simulation results and implementation results are obtained to determine the efficacy of the SAPF with ADALINE based neural network control for load compensation. SAPF system meets IEEE Standard 519 recommended harmonic standards for different rated nonlinear loads under balanced supply conditions.

Keywords: *Shunt Active Power Filter, Adaptive Linear Element, Hysteresis Current Control, PI Controller.*

1. Introduction

The proliferation of nonlinear load such as Thyristors, diodes degrade the P quality in the AC distribution network. Due to these issues, reactive P disturbance, weak PF, current harmonic generation, heating, the overall efficiency of the system and imbalance has developed in the AC transmission network. In olden days, the Passive filters (Pf) are used to alleviate harmonics, but the restriction of Pf for compensating has made active filters smart. This harmonic current causes V distortion and current distortion in the system, which can result in additional losses and measurement errors. Traditionally, Pf has used for compensating reactive P and for attenuating the harmonic distortion, but Pfis bulky, detained with aging, and it can resonate with the help of the supply impedance. The active P filters are the best tools for the compensation not only for current harmonics caused by the distortion of loads but also it applicable for reactive P, and the unbalance of nonlinear and fluctuating loads (Angelo Baggin), Nowadays, the SAPFs developed for compensating the harmonic distortion and reactive P requirement simultaneously. The Active Power Filter (APf) can connect in either series/shunt combination/ the combinations of both, which make unified P quality conditioners and hybrid configurations. The Shunt Active Filter (SAF) widely used than the series active filter due to the industrial applications, which require the current harmonic's compensation, and the active filters are very small, more selective, more versatile, and less chance of failure for component drift than its passive counterpart. The performance of SAPF definitely depends on load. The nature of load shall influence the shape of the AC supply current waveform. Therefore, the load definitely has an effect on the performance of APF. APF operate at slightly better efficiency at higher loads compared to lightly loaded condition. They studied widely and made development in theory and the application of APfs. Figure 1. Shows that he SAPF has two significant parts, which reference current extraction and PWM current controller. In SAPF, the reference current extraction obtained from the distorted line current and the PWM current controller that generates the switching patterns for VSC.

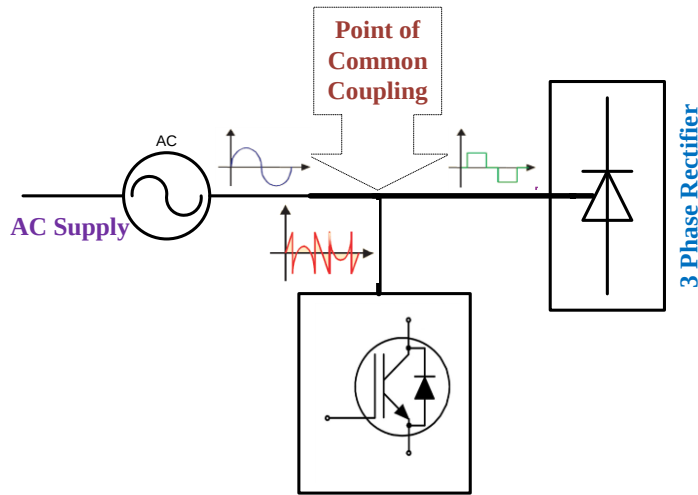


Figure 1. Model of SAPF

There are several current control techniques proposed for APF inner current control loops, such as Sinusoidal Pulse Width Modulation, Triangular Current Controller, Periodical Sampling Controller, and HCC. Hysteresis band PWM is basically an instantaneous feedback current control method of PWM where the actual current continually tracks the command current within a hysteresis band. The Hysteresis Current Controller (HCC) method [1] widely used by the researcher due to simple implementation and stability. In this paper, HCC has developed and analyzed, which produces the PWM pulses, and given to the APf. The performance of SAPF correctly based on the features of the current Control Algorithm (CA) and controllers. However, usually, one control scheme is more convenient for a few situations, but not for all situations. Adaptive linear element based current control system of SAPF has proposed for harmonic elimination, balancing of nonlinear loads, and PF correction. This paper originates with a short-term introduction to P quality problems.

2. Power Quality Issues

The Power Quality (PQ) problems are significant issues in the distribution system [2]. The PQ experts revealed that 50% of PQ problems occur due to grounding, neutral to the ground V, and ground current. The connected electrical equipment may fail due to these PQ issues. The causes of PQ classified into two types. The first category is natural causes, and the second type is human made causes. The natural cause occurs due to the lightning strikes on the transmission/distribution lines, which fails in the equipment. The human made causes occur due to the Energization of the transformer, capacitor switching, switching on or off the hefty loads, P electronic loads such as UPS, Adjustable Speed Drives. The variation of supply V which does not exceed one minute is called short duration V variation. These variations caused by faults and large load energization. The short duration V variations classified into

three categories, i.e. V sags, V swells, and interruptions. The V sag is the decrease in the fundamental frequency of the supply V for a short duration. The duration V sag varies between 5 cycles to one minute. The heavy load Energization may also cause the V sags. The V swell is the increase fundamental frequency of the supply V for a short duration. V swell is different from the V sag. The V sags and swells produced during the transformation of loads from one P source to another. The distortion in waveform occurs due to the deviation of the V waveform or current waveform from the fundamental sine wave. It can be classified as DC offset harmonics and notching. The DC offsets because of geomagnetic disturbances and half-wave rectification. They caused by poor grounding. The P electronic loads such as UPS, Adjustable Speed Drives can cause the harmonics in the P system. There is a huge loss in reactive power due to transmission over long transmission lines, and hence capacitor banks are used at various stages to inject reactive power into the system. The power quality issues are also a major problem for Power Distribution Systems.

The Total Harmonic Distortion (THD) [3] is the measurement of harmonics in the signal, and it should be less than 5%. During the operation of P electronic converter, the current commutates from one phase to another, which causes the periodic V distortion and it is called notching. During this period, the short circuit occurs between two phases resulting in V distortion [4]. During the connection of nonlinear loads, these problems have risen, and the harmonics are present in the output. The SAPF plays a vital role for harmonic elimination and in improving real P injection to the grid. Even though IRPT DFT based SAPF, SRF, SCD produced excellent results, and during a sudden change in load condition, it is not active. But ADALINE based SAPF system produces a good impact during a sudden change in load conditions and for cancellation of even and odd harmonics, improving PF and reducing reactive P. Since only the fundamental harmonic is considered to be desired, the rest can be considered as not desirable losses.

3. ANN control for SAPF System

The Artificial Neural Network (ANN) is a machine learning approach, and it consists of a large number of artificial neurons. The neurons in the ANN have less connection than the biological neurons. A large number of inputs given to each neuron in ANN. The output value of the neuron obtained by applying an activation function. There are several attempts used to control ANN. The ANN used for training the Neural Network by modifying the weights. The knowledge about learning tasks given in the form of examples called training examples. There are three types of network architectures, Single layer feed forward, multilayer feed

forward, and recurrent. This article proposes a new control strategy for III-Phases APF based on neural networks [6].

The SAPF connected with the III-Phase AC source. The gating signal for the APF generated by the PWM technique. To compensate nonlinear current in the system, the SAPF recycled. The operation of SAPF requires an III-Phase supply to provide the real P required to the load and losses in the SAPF. The PI controller is used to eliminate SAPF losses. Moreover, the output of the PI controller added to the weights, which are calculated by the ADALINE to maintain the DC voltage bus SAPF. During its operation, the AC mains are used to supply the real P needed for the load and switching losses of devices, losses in the reactor, and dielectric losses of the DC capacitor in the SAPF. The multiple harmonic optimization problem for both the control methods is formulated in such a way that for the whole selected range of harmonics a single control gain is searched. The reference source currents used to decide the APF switching has two parts: one is the essential occurrence component of the load current, which is being extracted using ADALINE, and the other corresponds to the losses in the APF, which is estimated using a PI controller over the DC V of SAPF [5]. To maintain the DC bus V of the SAPF, the output of the PI controller added to the weight calculated by the ADALINE. Finally, Continuous functions can be approximated by ANN effectively, which is proved both in theory and practice, as well as the discrete version of these continuous functions. These functions are common in production scheduling, resource allocation and so on.

3.1. Load Current Extraction Using ADALINE Network

Regardless of the complexity involved, the input-output relations with acceptable accuracy can be approximated using ANN [7] [8]. In P system applications, this is an additional ability of parallel computation and addictiveness to external disturbances, which has made ANN work with promising approximation techniques. The ADALINE and feed-forward MNN are the most commonly used ANN structures for fundamental/harmonic extraction. In this article, the implementation of ANN techniques in the P system has explained in detail. For practical illustration, PQ enhancement using shunt, APF, is considered. The basic concepts of ADALINE discussed in detail in the following subsections.

3.2 ADALINE Structure-based SAPF

A two-layer neural network consisting of an input layer and output layer is known as ADALINE. The functions with linear input-output relations can be estimated and can work reasonably even with some nonlinear real-world applications. The inputs (x_i) are multiplied

with modifiable weights (W_i) and then summed by a summation unit (output neuron). A1 and A2 are the input neuron that imparts the signals and B is the output neuron that receives the signals. The input neurons namely A1 and A2 are associated with the output neuron B over interdependent weights ($W1$ and $W2$). The estimated output (y) associated with a target (t_g), and the resulting error (e) is used by the LMS learning algorithm to train the network weights [3]. The ADALINE basic structure projected in figure 2.

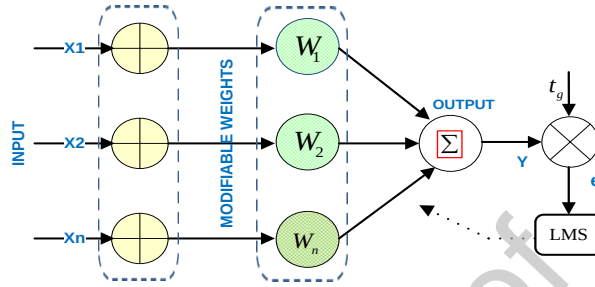


Figure 2. Adaline Structure

To realize the utilization of ADALINE for the abstraction of fundamental current magnitude in nonlinear load current, the distorted signal under load current (I_L) in Fourier series can be given as

$$I_L(t) = \sum_{k=1,2}^{\infty} a_k \sin(2\pi k f_0 t) + b_k \cos(2\pi k f_0 t) \quad (1)$$

F_0 is the fundamental supply frequency; a_k and b_k are the k^{th} harmonic sine and cosine amplitudes, respectively. From (1), the sine and cosine components easily estimated if the fundamental source frequency f_0 is known. However, the amplitudes a_k and b_k are to be determined. About the ADALINE formulation, the known functions interpreted as the inputs to the ADALINE, and the unknown amplitudes (a_k and b_k) considered as the network weights. The network output (y) is the assessed load current, and the measured load current (I_L) is the target (t_g). The assessed load current (y) as

$$y^n = \sum_{k=1,2}^k a_k^n \sin(2\pi k f_0 t) + b_k^n \cos(2\pi k f_0 t) \quad (2)$$

Where y^n sampled at $t_n = nTs$, k , is the harmonic index. H is the highest order harmonic. The coefficients a_k^n and b_k^n are the amplitudes of the K^{th} sine and cosine harmonic terms, respectively [6]. In equ (2), the even-order and triplen harmonics excluded due to the half-wave symmetry and III-Phase system configuration. The estimated load current (y^n) written in vector format as

$$y^n = (w^n)^T x^n \quad (3)$$

$$\text{Where } (W^n)^T = [a_1^n a_5^n \dots a_H^n b_1^n b_5^n \dots b_H^n] \quad (4)$$

$$(x^n)^T = [\sin(2\pi f_0 t^n) \sin(2\pi 5 f_0 t^n) \dots \sin(2\pi H f_0 t^n) \cos(2\pi f_0 t^n) \cos(2\pi 5 f_0 t^n) \dots \cos(2\pi H f_0 t^n)] \quad (5)$$

The weights realized by using the LMS algorithm, which built on the gradient descent optimization procedure. To reduce the error between the targets $tg^n = I_L^n$ for the resulting outcome (y^n), the gradient descent applied concerning the network weights. Thus the weight update rule is [9]

$$W^n = W^{n-1} - \eta \cdot \nabla E^n(w) = W^{n-1} + \eta \cdot e^n \cdot x^n \quad (6)$$

Where η is the step size or learning rate of the weight update. The activation function is selected as tan sigmoid for the input layer and positive linear activation function for target layer.

4. Control Algorithm

During the operation of the SAPF system [10], the real P needed for the load and some losses in the SAPF complete by AC mains. Therefore, the location source currents that are used to decide the switching of the SAPF have two parts, one is the real fundamental frequency component of the load current, and the other corresponds to the losses in the SAPF.

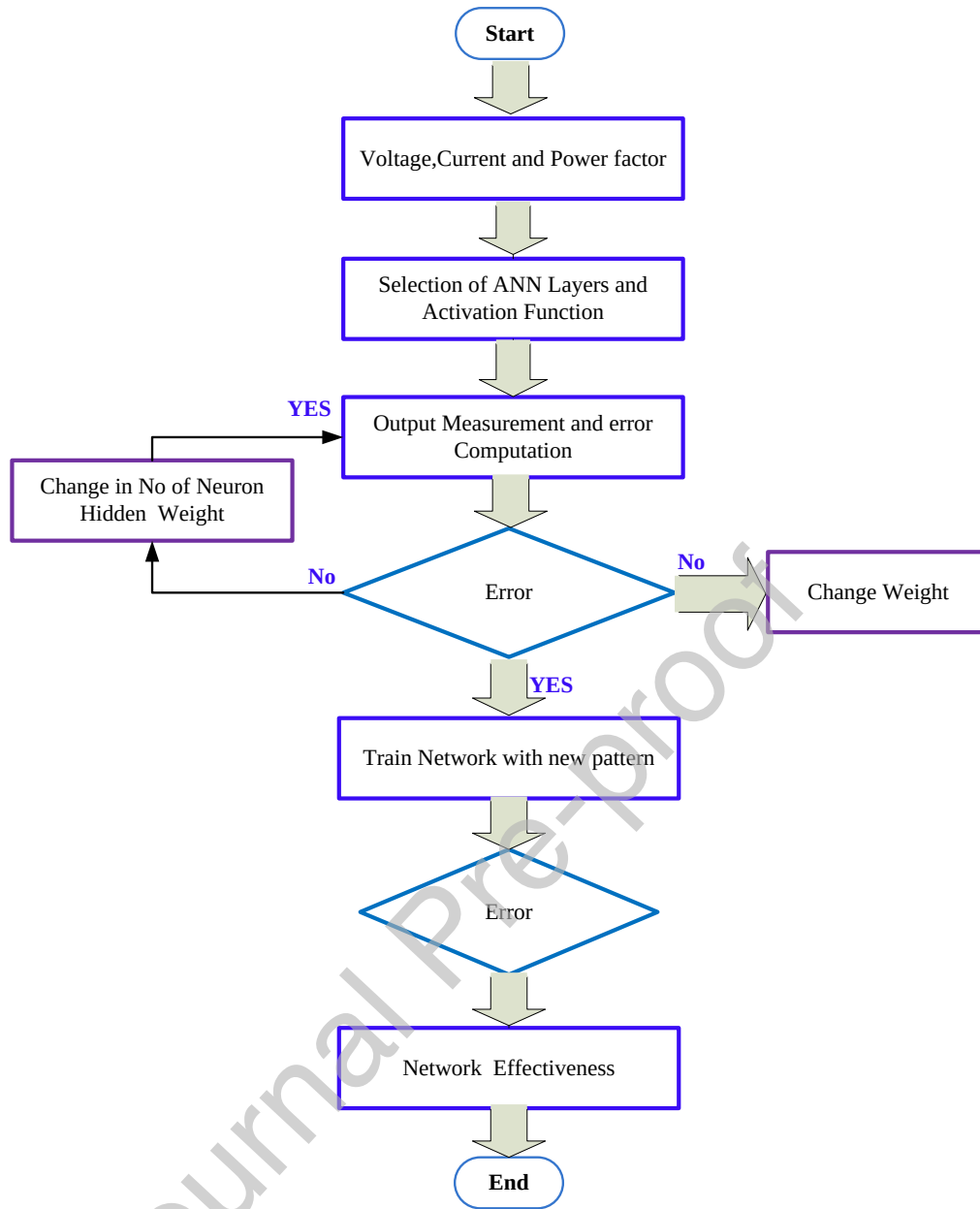


Figure 3. ANN Flow Chart for reference Current generation

To preserve the DC bus V of the SAPF, the production of the PI checker is additional to the weight intended by the ADALINE and compared with fuzzy based PI Controller [11] shown in figure 3.

4.1. Fundamental Frequency Current from Load Current

The basic theory of the proposed method based on the LMS algorithm, and its training through ADALINE tracks the unit vector prototypes to maintain minimum error. The load current consists of real and reactive components of a positive and negative sequence, and harmonic components presented. The CA based on the extraction of the current component in phase with the V. The unit V should be in phase with the system V same unit amplitude to estimate the fundamental frequency positive real sequence component of the load current.

Phase V of the system provides the unit V pattern and it can be denoted as the instantaneous values at PCC to the V, So Point of common coupling, $I_{load} + I_{source} + I_{filter} = 0$, But this sum is vectorial, not in magnitude. For effective filtering, I_{filter} should be anti-phase (180 degrees apart for each harmonic frequency) with I_{load}

$$V_u = v_{pcc}/V \quad (7)$$

For proper estimation of load current components, the unit V pattern understated. The unit vector patterns are generated by the sampled phase V which is filtered by the second-order Butterworth filter at the rate of 50Hz. For a single-phase, the active P component of the load current computed by increasing weight with the unit pattern, and this

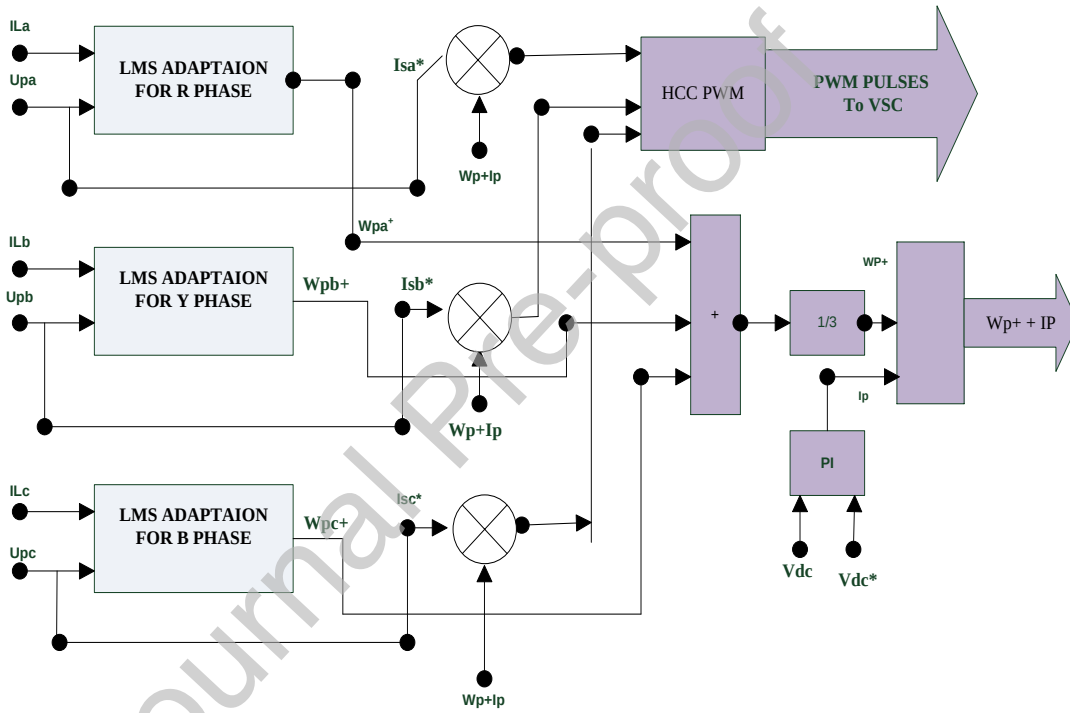


Figure 4. Reference current generation using the LMS adaptation algorithm

weight (W_p) is estimated using the Adaline technique is shown in figure 4. This weight is variable and variations as per the load current and the magnitude of phase V. The scheme for computing weights corresponding to the fundamental frequency real component of the load current, based on the LMS algorithm tuned Adaline technique [12], tracks the unit vector template to maintain minimum error. The estimation of weight given as per the following expression.

$$W(n+1) = w(n) + \mu E\{u_n \times e^*(n)\} \quad (8)$$

The rate of junction and accuracy of valuation decided by the value of μ (convergence coefficient). The practical range of convergence coefficient lies 0.01 and 1.0. The higher value of μ raises the rate of convergence of weights [13], but it leads to erroneous results, whereas the low value of μ increases the accuracy of estimation but leads to a slow convergence of weights. The best trade-off observed in accuracy and the rate of convergence with μ equal to 0.2. The definite sequence of load current determined by the III-Phase unit pattern with the average equivalent weight shown in figure 3.

$$W_{Rp}^+ = \frac{W_{Pa}^+ + W_{Pb}^+ + W_{Pc}^+}{3} \dots\dots\dots(9)$$

For proper reference current estimation, the weights are averaged to compute the equivalent weight for definite sequence current components in the decayed form. The averaging of weights aids in removing the unbalance in the current components. The error signals are determined by obtaining the difference between the calculated and targeted output that measures in the output layer.

5. PI Controller for Maintaining Constant DC Bus V of SAPF

To calculate the second component of reference active current, a reference DC bus V compared with the sensed DC bus V of SAPF. The comparison of sense DC bus V to the reference DC bus V of VSC of SAPF results in a V error, which in the n^{th} sampling instant expressed as:

$$V_{dc}(e) = V_{dc}^* - V_{dc}(n) \dots\dots\dots(10)$$

The PI controller processes the error signal $v_{dc}(n)$ and the output $I_p(n)$ at the n^{th} sampling instant expressed as

$$I_p(n) = I_{p(n-1)} + K_{idc}\{V_{dc}(e) - V_{dc}(e)_{(n-1)}\} + K_{pdc}V_{dc}(e) \dots\dots\dots(11)$$

Where K_{pdc} , K_{idc} are the proportional and integral gains of the PI controller.

The output of the PI controller accounts for the SAPF losses, and it considered as a loss component of the current, which is new with the weight and estimated using the Adaline corresponding to the fundamental frequency positive sequence reference dynamic load current component. These currents considered as a reference source currents $I_{ref}(I_{sa}^*, I_{sb}^*, I_{sc}^*)$, and along with the detected source currents $I_{act}(I_{sa}, I_{sb}, I_{sc})$, these are fed to the HCC PWM controller to control the source currents to follow the reference currents. For triggering the PWM generator and with design of passive element for producing appropriate wave shape that cancels out the source voltage harmonics. This current control technique results in

control of the source currents and therefore requires less mathematical Calculations [14] [15]. The LMS algorithm is used to compute weights. The update equation of the weights based on the LMS algorithm described in (3) for each phase. The structure of such ADALINE control shown in Figure 2. The weights are averaged not only for averaging at the fundamental frequency and also to cancel out the sinusoid ally wavering components in weights present due to harmonics in the LC. The averaging of weights in different phases shown in Figure 3. The control circuit generates the sine reference signal wave of desired magnitude and frequency, and it is compared with the actual signal. As the signal exceeds a prescribed hysteresis band, the upper switch in the half-bridge is turned OFF and the lower switch is turned ON. As a result the output transits from +0.5Vdc to -0.5V dc , and the signal start to decay. As the signal crosses the lower limit, the lower switch is turned OFF and the upper switch is turned ON. The actual signal wave is thus forced to track the sine reference wave within the hysteresis band limits.

Thus, Adaline trained at the fundamental frequency of a particular sequence in phase with the Voltage. The second harmonic ripple is produced in the DC bus Voltage because of the unbalance in LC. This ripple may cause the harmonic component in the source currents. So it has to be filtered out before feeding the signal to the PI controller. For this purpose, the DC bus Voltage is filtered using an LPF. Since the critical volume of the reference current is calculated using an Adaline-based extractor, the effect of the delay produced by the LPF is negligible in practical cases [16]. The loss in load power due to the insertion of a component or device at some point in a transmission system; generally expressed as the ratio in decibels of the power received at the load before insertion of the apparatus, to the power received at the load after insertion.

5.1 Role of DC Capacitor

The DC link capacitor mainly serves two purposes i) It maintains almost a constant DC voltage ii) It serves as an energy storage element to supply real power difference between load and source during transients. In steady state the real power supplied by the source should be equal to the real power demand of the load plus some small power to compensate the losses in the active filter. Thus the DC link voltage can be maintained at a reference value. However, when the load condition changes the real power balance between the mains and the load will be disturbed. This real power difference is to be compensated by the DC link capacitor. This changes the DC link voltage away from the reference voltage. In order to keep the satisfactory operation of the active filter the peak value of the reference current must be

adjusted to proportionally change the real power drawn from the source. This real power charged/discharged by the capacitor compensates the real power consumed by the load. If the dc link voltage is recovered and attains the reference voltage the real power supplied by the source is supposed to be equal to that consumed by the load and also the losses. Thus, the peak value of the reference source current can be obtained by regulating the average voltage of the DC link capacitor. A smaller DC link voltage than the reference voltage means that the real power supplied by the source is not enough to supply load demand. Therefore, the source current (i.e. the real power drawn from the source) needs to be increased. While a larger DC link voltage than the reference voltage tries to decrease the reference source current. As the DC link capacitor is used as an energy source, the output of the inverter is the voltage that has to be filtered by an inductance or a high order filter to limit the level of the ripple current. The filter inductor is used to attenuate the ripple of the inverter current caused by the switching of the inverter. Hence the design of filter inductor is based on the principle of harmonic current reduction

6. Simulation Setup

The nonlinear load modelled by an 3Phase diode bridge converter connected to AC mains. At the DC bus of the diode bridge converter, a Resister inductor filter combination is employed. A detailed control scheme is shown in Figure 4. The control system with 3 phase Adaline blocks and a PI controller to preserve the DC bus Voltage of SAF [17]. The flow of signals are such that the sensed LC fed to the Adaline block, which estimates the weight corresponding to the real fundamental frequency part of the LC. The loss portion of the reference current is

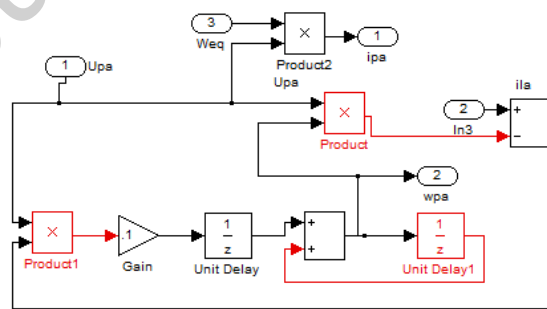


Figure 5. Adaptive Control for phase “A.”

Figure 5 shows the MATLAB model of the SAPF system. The nonlinear load is using an III-Phase diode bridge converter connected to AC mains. A detailed control scheme

presented in figure 5. The control system with III-Adaline blocks and a PI controller to keep the DC bus V of SAPF.

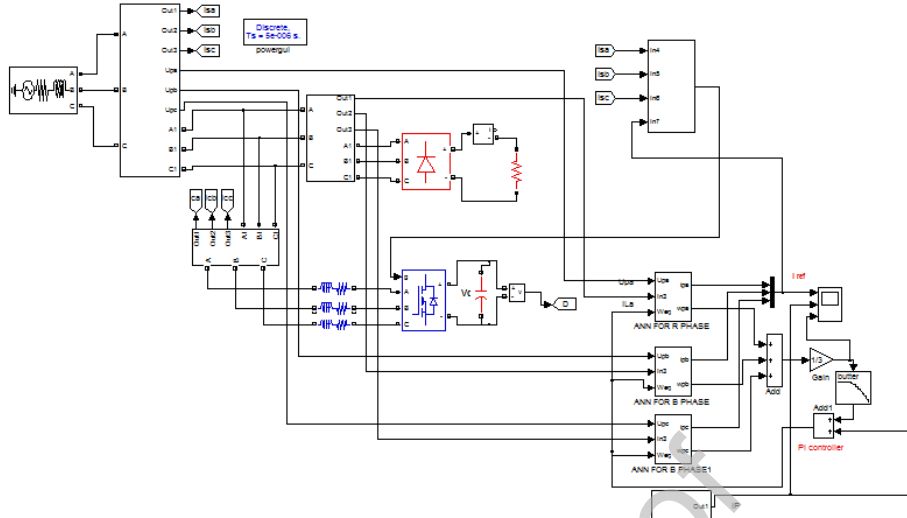


Figure 6. Simulation Setup

Figure 6 shows that the source current has deviated from actual sinusoidal current and harmonics offered in the output. Harmonics results in weak PF, P unbalance, and increasing reactive P. Finally, the quality of P has been degraded. The linear adaptive neural method has generated the required.

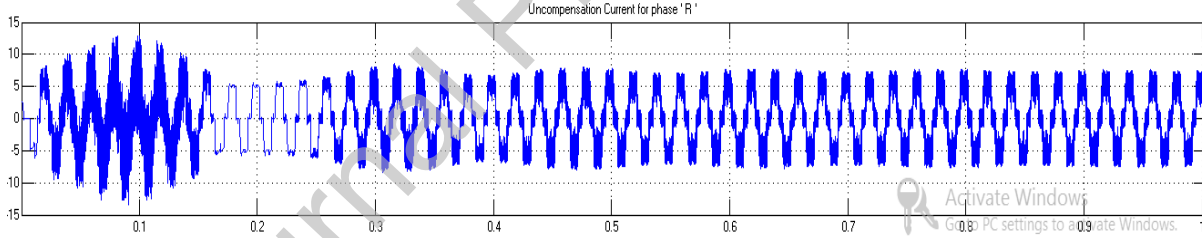


Figure 7. Uncompensation Source Current "R" Phase

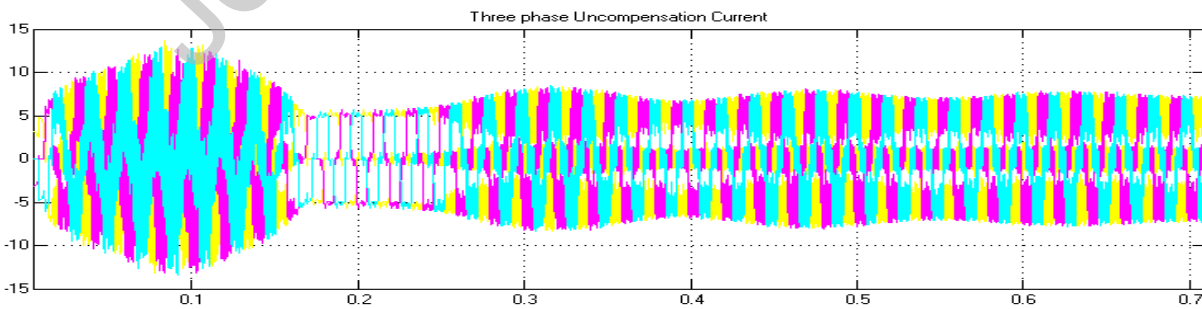


Figure 8. Uncompensation Current

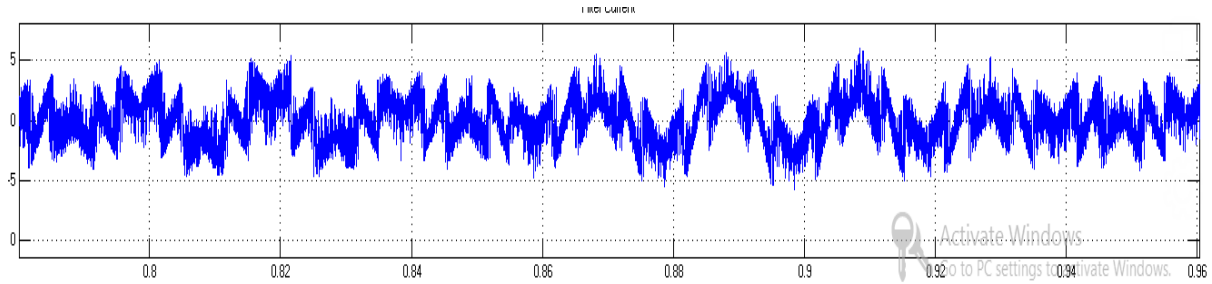


Figure 9. Filter Current

The figure 10 shows that the filter current added into PCC; the source current is normalized and in phase with the source Voltage. The figure 11 describes about 3-Phase source current, so the PF of the overall system has improved from 0.87 to 0.981.

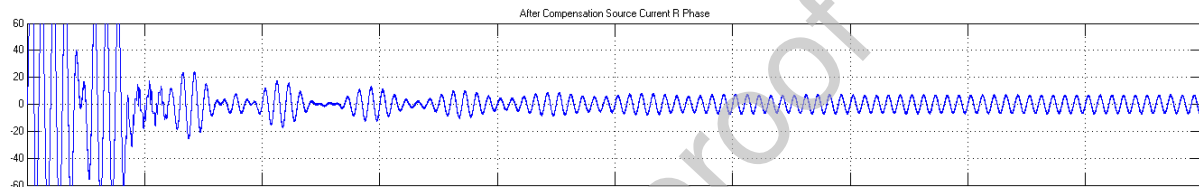


Figure 10. Source Current after Compensation

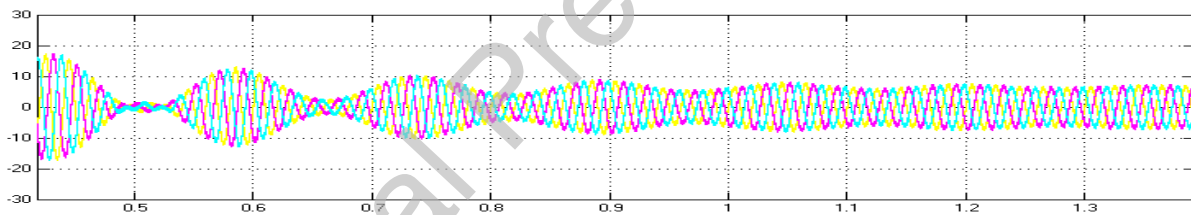


Figure 11. 3-Phase Source Current after Compensation

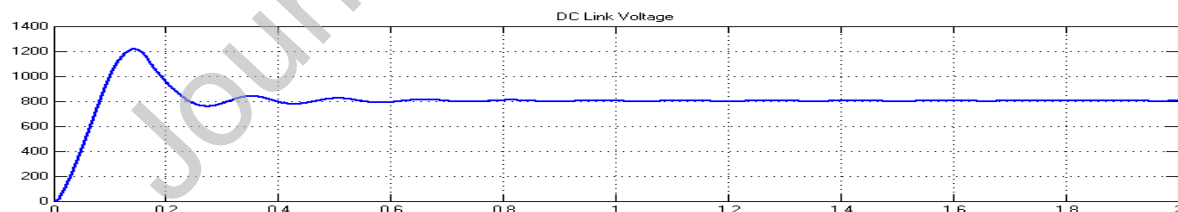


Figure 12. DC Link V

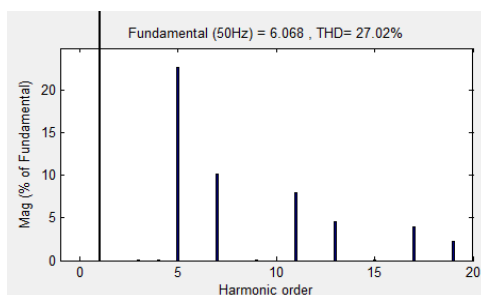


Figure 13. Before Compensation THD

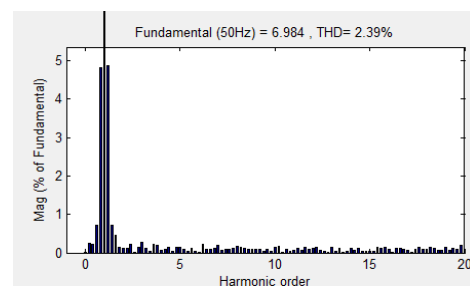


Figure 14. After Compensation THD

When compared to IRPT based DFT, the Synchronous reference frame method, the THD of the proposed method reduced from 27.02% to 2.39%. Figure 12 shows the THD, PF of the system, and the DC link settling time of the system. On comparing with all the advanced methods, the ANN-based SAPF system produces excellent results, and the robustness of the system is also higher. The ANN Based SAPF system Input Current and Voltage with different Load as shown in table 1.

Table 1. SAPF Input current and voltage at different Resistive loads.

Voltage-RPhase	Current - Rphase	Input PF	Power –R Phase	Output Current-R Phase	Output Voltage-R Phase	Output Power in Watts	Efficiency in R Load
230	0.79	0.96	172	0.51	550	350	49
231	1.28	0.95	290	1.01	587	550	53
230	2.56	0.95	585	1.87	575	1100	53
232	3.12	0.96	884	2.52	567	1650	54
232	4.51	0.97	978	3.72	591	2200	44
231	5.47	0.98	1150	4.79	569	3100	37

The ANNs learning capabilities to adaptively choose the power system parameters for both to compute the reference currents and to recharge the capacitor value requested by V_{DC} voltage in order to ensure suitable transit of powers to supply the inverter. Figure 13 and figure 14 shows that harmonics profile for before and after compensation. Table 2 shows that the simulation parameter for ADALINE based SAPF.

Table 2. Shunt APF system data for the simulation studies.

III-Phases balanced at 415V RMS, 50Hz	
Source	Source Impedance $R=0.1\ \Omega$, $L=0.05e^{-1}$ DC Link capacitors: $1500\ \mu F$ Filter Resistance $0.1\ \Omega$, Filter Inductance $=4.35e^{-3}$ DC Link Reference: 800 PWM = HCCB=0.01
SAPF System	DC LINK Proportional Gains $K_p=0.4$ Integral gains $K_i=1$
II nd Filter Cut off Frequency = 25Hz	
Damping factor = 0.707	
3-phase diode rectifier with $L_{in}=500\ \Omega$	
Load	DC side load- $R=100\ \Omega$

For analysis, the factors which are harmonic distortion, DC Link settling time, and PF of the system have compared. Figure 15. Shows that ADALINE based SAPF system produced.



Figure 15.TH D

Better results when compared with all algorithms based SAPF systems. Figure 16 and figure 17.shows the settling time of the capacitor and PF of the system compared. All three parameters have noticed and compared, which has shown in figure 18.

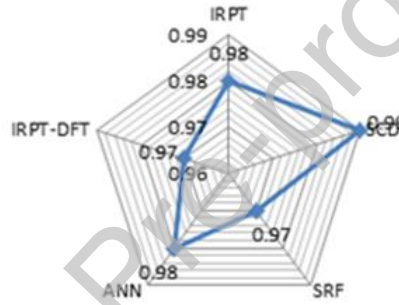


Figure 16.Power factor

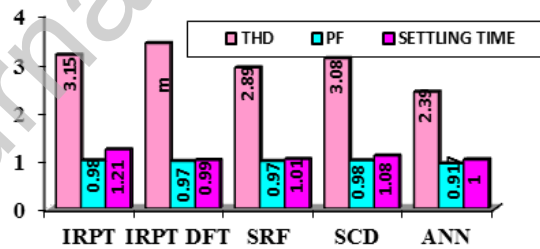


Figure 17.Over all Comparison

A three phase diode rectifier with an RL load was used as a harmonic producing load. The rectifier load value is (resistance was $10/3 \Omega$ and the inductance 60 mH) as shown in table 3.

Table 3. Harmonic Supply Current Phase A

Harmonic Supply Current Components			
$I_{sa(n)}/I_{sa}$			
n	load	Pq	ANN
5	19.59	0.37	0.28
7	13.56	0.43	0.37
11	8.06	0.17	0.04
13	6.48	0.27	0.24
17	4.38	0.41	0.06
19	3.63	0.21	0.17
23	2.51	0.11	0.06
25	2.08	0.15	0.12
29	1.43	0.21	0.04
31	1.18	0.05	0.08
35	0.82	0.07	0.04
37	0.70	0.03	0.05
41	0.56	0.16	0.03
43	0.51	0.01	0.05
47	0.46	0.11	0.04
49	0.44	0.05	0.05
THD	26.91	1.05	0.74

In IRPT method requires many mathematical calculations that necessary for evaluating compensation current, and IRPT based DFT method gives improper synchronization, leads to poor results.

7. Conclusion

The analysis, simulation, and implementation of a new three-phase SAPF consisting of a three-leg VSC with III-Phase rectifiers, controlled by NN algorithm have been carried out, and its performance has demonstrated for harmonic current elimination along with reactive P. The dc bus V of the III-Phase SAPF has been regulated to the reference dc bus V under constant load. The Adaline based CA for the SAPF has demonstrated under stable operating conditions. The operation of a SAPF has been observed to be entirely satisfied with the unstable and nonlinear loads. The minimalism of the control and the inherent linearity of the Adaline based control are some of the salient features of the CA. Moreover, it is clear from the offered outcomes that a THD of as low as 2.39 % achieved for V source fed type of nonlinear loads. From these simulated, it concluded that a SAPF controlled by the Adaline technique could provide reactive P compensation, harmonic mitigation, and balancing of unbalanced and nonlinear loads.

Conflict of Interest

This paper has not communicated anywhere till this moment, now only it is communicated to your esteemed journal for the publication with the knowledge of all co-authors.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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